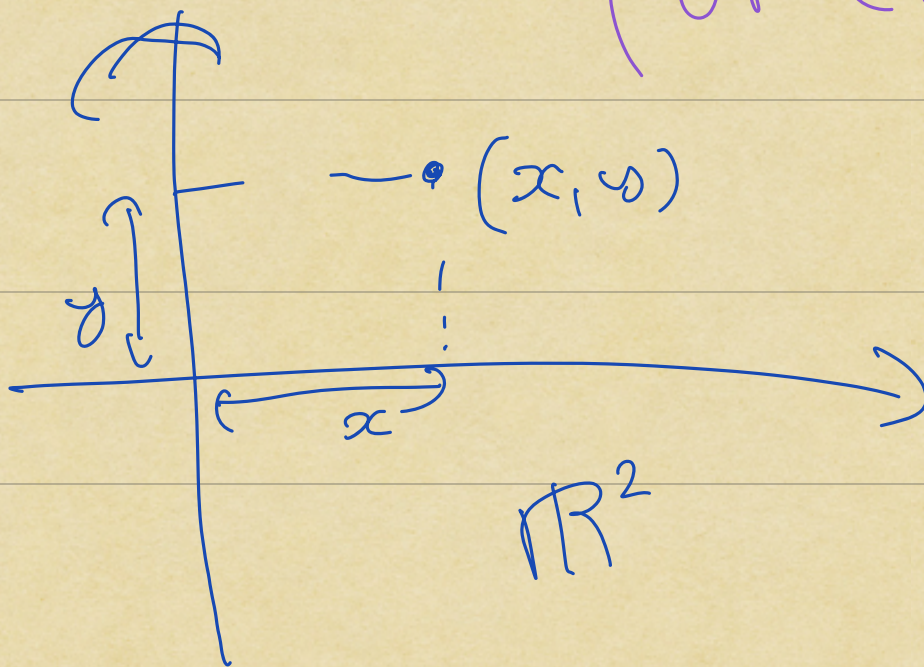
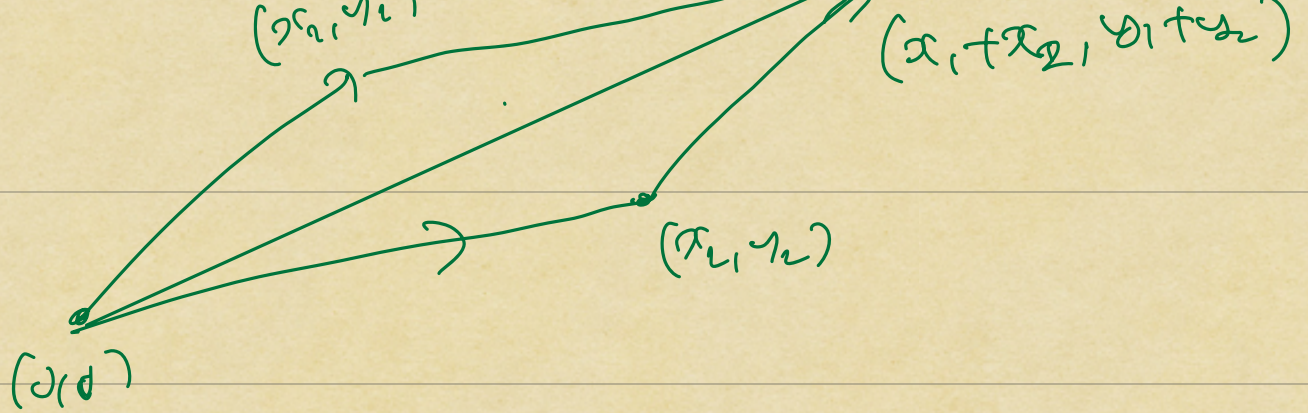


Examples

① F is a vector space over F

② $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$
is a vector space over $\mathbb{R}$
(or even over $\mathbb{Q}$)





(3) V, W : two F -vector spaces

$$V \times W = \{ (v, w) : v \in V, w \in W \}$$

is also an F -vector space

with

- $0_{V \times W} = (0_V, 0_W)$

- $(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2)$

- $a \in F, a \cdot (v, w) = (a \cdot v, a \cdot w)$

④ For any field F , $n \in \mathbb{Z}_{\geq 0}$, we get an F -vector space

$$F^n = \underbrace{F \times F \times \dots \times F}_{n\text{-times}}$$

Notation: A vector in F^n will be written in the form of a column
(instead of $(a_1, a_2, \dots, a_n)$)

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, a_i \in F$$

(5) F : field

$$\text{Poly}(F) = \left\{ a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \right\} \\ \left\{ : a_i \in F, n \in \mathbb{Z}_{\geq 0} \right\}$$

This is the set of

polynomials with
coeffs in F

$$(\text{zero}) \cdot 0 = 0$$

$$\begin{aligned} (\text{Addition}) &= (a_n x^n + \dots + a_1 x + a_0) \\ &\quad + \\ &\quad (b_n x^n + \dots + b_1 x + b_0) \\ &= \frac{(a_n x^n + \dots + a_1 x + a_0) + (b_n x^n + \dots + b_1 x + b_0)}{} \\ &= (a_n + b_n) x^n + \dots + (a_1 + b_1) x + (a_0 + b_0) \end{aligned}$$

$$\begin{aligned} (\text{Scalar mult.}) &\cdot r \in F \\ &r \cdot (a_n x^n + \dots + a_1 x + a_0) \end{aligned}$$

$$= (r \cdot a_n) x^n + \dots + (r \cdot a_1) x + r \cdot a_0$$

⑥ Fix $m \in \mathbb{Z}_{\geq 0}$

$$\text{Poly}_{\leq m}(F) \subseteq \text{Poly}(F)$$

|| is an
 $\left\{ \begin{array}{l} \text{polynomials} \\ \text{of deg} \leq m \end{array} \right\}$ F -vector
 subspace

E.g. $\text{Poly}_{\leq 0}(F) = \{a_0 : a_0 \in F\}$

$$\text{Poly}_{\leq 1}(F) = \{a_1 x + a_0 : \begin{array}{l} a_1, a_0 \\ \in F \end{array}\}$$

$$\text{Poly}_{\leq 2}(F) = \left\{ a_2 x^2 + a_1 x + a_0 : \begin{array}{l} a_0, a_1, a_2 \in F \end{array} \right\}$$

Defⁿ (Subspace)

V : F -vector space

$W \subseteq V$: subset

W is an F -vector subspace

(or simply subspace)

if: (1) (closure under $+$)

$$v_1, v_2 \in W \Rightarrow v_1 + v_2 \in W$$

(2) (closure under $\cdot$)

$$a \in F, v \in W$$

$$\Rightarrow a \cdot v \in W$$

(3) (non-emptiness)

$$W \neq \emptyset$$

Rmk: (3) is equivalent
to asking for $0 \in W$.

Basic properties of
vector spaces

$$(1) \quad v \in V, \quad 0 \cdot v = 0$$

pp:

$$\begin{aligned} 0 \cdot v &= (0 + 0) \cdot v \\ &= 0 \cdot v + 0 \cdot v \\ &\quad 1 + (-0 \cdot v) \end{aligned}$$

$$\begin{aligned} 0 &= (0 \cdot v + 0 \cdot v) + (-0 \cdot v) \\ &= 0 \cdot v + (0 \cdot v + (-0 \cdot v)) \end{aligned}$$

$$= 0 \cdot v + 0$$

$$= 0 \cdot v$$

(2) $v \in V, 1 \cdot v = v$ (Has to be imposed)

(3) $v \in V, -1 \cdot v = -v$

Need to show:

$$v + (-1 \cdot v) = 0$$

||

$$1 \cdot v + (-1) \cdot v$$

||

$$(1 + (-1)) \cdot v = 0 \cdot v$$

✓
QED

(4) $a \in F, a \neq 0$

$$a \cdot (a^{-1} v) = (a \cdot a^{-1}) \cdot v$$

$$\begin{aligned} &= 1 \cdot v \\ &= v \end{aligned}$$

PF of R_{mk}

If W satisfies (1) & (2)
of the defⁿ of a subspace

$$W \neq \emptyset \Rightarrow 0 \in W$$

$\Downarrow$

$$\exists v \in W \Rightarrow 0 \cdot v = 0 \in W$$

$\Downarrow$

$$-1 \cdot v \in W \Rightarrow -v \in W$$

$$\Rightarrow v + (-v) \in W$$

$$\Rightarrow 0 \in W.$$

Rmk (2)

V : F -v. space

$W \subseteq V$: subspace

Then W is an F -v. space
in its own right.

(7) (Non-example)

$\text{Poly}_m(F) \subseteq \text{Poly}(F)$

$\left\{ \begin{array}{l} \text{Polynomials} \\ \text{of deg} = m \\ a_m x^m + \dots + a_1 x + a_0 \\ a_m \neq 0 \end{array} \right\}$

Doesn't
contain 0.

$$\text{Poly}_m(F) \cup \{0\}$$

"union"

ex. $\text{Poly}_2(F) \cup \{0\}$

$$= \{a_2x^2 + a_1x + a_0 : \text{Either } a_0 = a_1 = a_2 = 0 \text{ or } a_2 \neq 0\}$$

$$\underbrace{x^2 + 1} + \underbrace{(-x^2)} = 1$$

$$\in \text{Poly}_2(F)$$

~~$\notin$~~

$$\text{Poly}_2(F) \cup \{0\}$$